

AI2 Exercise 2b

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Answer Set Programming

Task A For Task A, we defined several rules: `adjacent/4` takes two rows and cols as input and can be derived if the two positions on the grid are horizontally, vertically or diagonally adjacent. `allowed/2` takes a row and col as inputs and can be derived if the row and col are adjacent to a wall. Then we placed light bulbs by “guessing” on allowed positions. We then added a constraint that every valid position has to be `litUp`, for this we defined the rules `litUp` and `blockedByRow/Col`. Result: clingo version 5.7.2 showed 801 correct models for the provided example instance of the grid.

Task B For Task B, we just added one constraint and rule: `adjacentToWall/4` is self explanatory, it can be derived if one position is a wall and the other position is adjacent to it. The constraint itself ensures with the aggregate function `count` that each wall doesn't have more than `k` light bulbs next to it. Result: clingo version 5.7.2 showed 44 correct models for the provided example instance of the grid.